COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 - Data Structures & Algorithms	**SECTION:**	CPE21S2
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NAME:	Acala Antonio Arididdon Astudillo Bandigan	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

ACTIVITY NO **Seatwork 9.1 Binary Search Tree**

1. PROCEDURE OUTPUT

1. What is a binary search tree?

A binary search tree is a type of hierarchical, node-based data structure that keeps its elements organized in a sorted order to enable rapid searching, insertion, and deletion. A binary search tree uses a parent-and-child based structure where the left child contains a value smaller than the parent's, while the right child contains a value greater than the parent's, typically with no duplicate values.

2. Where can binary search tree be used?

They can be used for fast searching, inserting, and deleting data, such as in databases, dictionaries, and search systems.

3. What is a tree traversal?

Tree traversal is the process of visiting or going through every node in tree. We follow a specific order depending on what we want to do with the data. Some common types are pre-order, in-order, and post-order traversal, and each one visits the nodes in a different sequence.

4. Differentiate a post-order and pre-order traversal give examples.

Post-order traversal visits the children of the root node first. It follows a left-right-root policy, meaning it traverses the left child first, then right child, and finally the root node. Pre-order traversal does the opposite, it visits the root node first. It follows a root-left-right policy meaning it traverses the child nodes after the root node.

5. What is a parse tree and its use?

Parse trees is a structure that represents how a string is derived from the start symbol of a grammar. It can be used to represent real-world constructions like sentences or mathematical expressions. It consists of the root node which represents the start symbol (S), the interior nodes representing non-terminal symbols or variables (V), and leaf nodes representing the terminal symbols or the actual input.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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